



**Verified Carbon
Standard**

FLOATING SOLAR PROJECT OF GREENAM ENERGY



Document Prepared by EKI Energy Services Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Greenam Energy Private Limited (GEPL/ the Developer) is a special purpose vehicle formed by AMIH (AM International Holdings) to develop a 24.7MWp (22 MWac) floating solar PV plant (the Project) on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district in the Indian state of Tamil Nadu.

The purpose of this Project is to utilize the area of the reservoir to generate clean energy for captive use by the AMIH Group companies - Southern Petrochemical Industries Corporation Limited (SPIC) and Greenstar fertilizers (GFL). A common power purchase agreement (PPA) had been executed between GEPL and the off-takers, namely SPIC and GFL.

The project boundary includes the project power plant (project site where the solar power plant has been installed including the solar power plant, power evacuation infrastructure, energy metering points, switch yards and other civil constructions) implemented in the state and all other power plants/units connected physically to the national (Indian) grid that the project power plants is connected to.

The average annual generation of the project is 36,873 MWh. The emission reduction from the project is estimated to be around 36,783 tCO₂e /annum and a cumulative emission reduction of 258,117 tCO₂e for the entire crediting period of 7 years.

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Validation/ Verification	(DD-Month- YYYY-- DD- Month-YYYY)			
	<u>Year...</u>			
<u>Total</u>				

1.2 Sectoral Scope and Project Type

The project is falling into the sectoral scope 1 (Energy renewable/non-renewable sources),

sectoral scope 01: Energy industries¹

The project is not a grouped project.

ACM0002 Grid-connected electricity generation from renewable sources -- Version 21.0²

1.3 Project Eligibility

As per section 2.1.3 of VCS Standard ver-4.4, table no. 1 excluded project activities-

(The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems e.g., batteries).

Thus, the proposed project is eligible under the scope of the VCS program.

1.4 Project Design

The Project is not a Group project

Eligibility Criteria

Not Applicable

1.5 Project Proponent

Organization name	Greenam Energy Private Limited
Contact person	P.Senthil Nayagam
Title	Director
Address	GREENAM ENERGY PRIVATE LIMITED "SPIC HOUSE", 88 Mount Road, Guindy,, Chennai, Tamil Nadu, India, 600032
Telephone	
Email	senthilnayagam@spic.co.in

1.6 Other Entities Involved in the Project

¹https://cdm.unfccc.int/DOE/scopes.html?gclid=Cj0KCQiA_P6dBhD1ARIsAAGI7HDggTNy1T26sHqdoxGb1NHqGz8v6FDRT6Dt4M8vU4uw1i_B-umxt4aAsmDEALw_wcB

² <https://cdm.unfccc.int/methodologies/DB/HF3LP6O41YY0JIP1DK6ZRJO9RSCX3S>

Organization name	EKI Energy Services Ltd.
Role in the project	Project Representative
Contact person	Mr. Manish Dabkara
Title	MD & CEO
Address	Office No 201, Plot No 48, Scheme 78, Part 2, Vijay Nagar, Indore-452010, Madhya Pradesh
Telephone	+91 9589899649
Email	registry@enkingint.org , Sandeep.kurmi@enkingint.org

1.7 Ownership

The respective project commissioning certificate for project activity are the supporting documents to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.4) – “a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals”. Also, other legal compliances may be considered;

- Consent to Operate
- Commissioning documents
- Consent to Establish

1.8 Project Start Date

12-January-2022.

1.9 Project Crediting Period

Total Crediting period is 3*7 Years

Crediting period is from 12-January-2022 to 11-January-2029

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

When completing a draft project description for the purpose of listing on the pipeline as under development, complete the following information; otherwise, delete this text.

The estimated annual GHG emission reductions/removals of the project are:

☐ <20,000 tCO₂e/year

- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	37,352
Year 2	37,192
Year 3	37,032
Year 4	36,872
Year 5	36,714
Year 6	36,556
Year 7	36,399
Total estimated ERs	258,117
Total number of crediting years	7
Average annual ERs	36,873

1.11 Description of the Project Activity

The proposed 24.7MWp (22 MWac) floating solar PV plant shall be developed on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district in the Indian state of Tamil Nadu. Approximately 48.59acres of total water cover is expected to be available for

developing the proposed 22 MWac floating solar PV plant using a methodology ACM0002 ver-20.0³.

The power generated by the floating solar PV plant have evacuated to the utility substation at 110kV. Depending on the quantum of power injected into the substation as recorded by the custody transfer meter, the state utility i.e., TANGEDCO would allocate power to the two off takers based on prior agreements with GEPL. Wheeling charges shall be borne by the off takers.

The propose project activity is the installation of a new 22 MW AC Solar captive power plant/unit at the site where no renewable power plant was operating prior to the implementation of the project activity (green-field plant⁴)

Details of proposed solar power project is outlined in the table below:

Project Owner	Capacity of the Power plant	Date of Commissioning	State	Connectivity
Greenam Energy Pvt Ltd	22 MW _(AC)	12/01/2022 ⁵	Tamil Nadu	State Grid

The electricity generated from project activity (operational solar power plant) is connected to the state grid. The project activity thus reduces the anthropogenic emissions of greenhouse gases (GHGs) associated with equivalent amount of electricity generation from the existing grid connected power plants (mostly fossil fuel based) and from addition of new generation sources into the grid. The average annual generation of the project is 36,873 MWh. The emission reduction from the project is estimated to be around 36,873 tCO_{2e} /annum and a cumulative emission reduction of 258,117 tCO_{2e} for the entire crediting period of 7 years

1.12 Project Location

Greenam Energy Private Limited (22 MW) project activity is located in Muthiahpuram village Thoothukudi district in Tamil Nadu, with the geographical Coordinates of 8°44'6"N to 78°7'44"E

³ <https://cdm.unfccc.int/UserManagement/FileStorage/AG07ZJQ3EXD42LT5YV9HR16M8KINPO>

⁴ As per ACM0002 ver20.0, a Greenfield power plant is defined as a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity

⁵ Date of Grid synchronization of the project 22MW is 12/01/2022

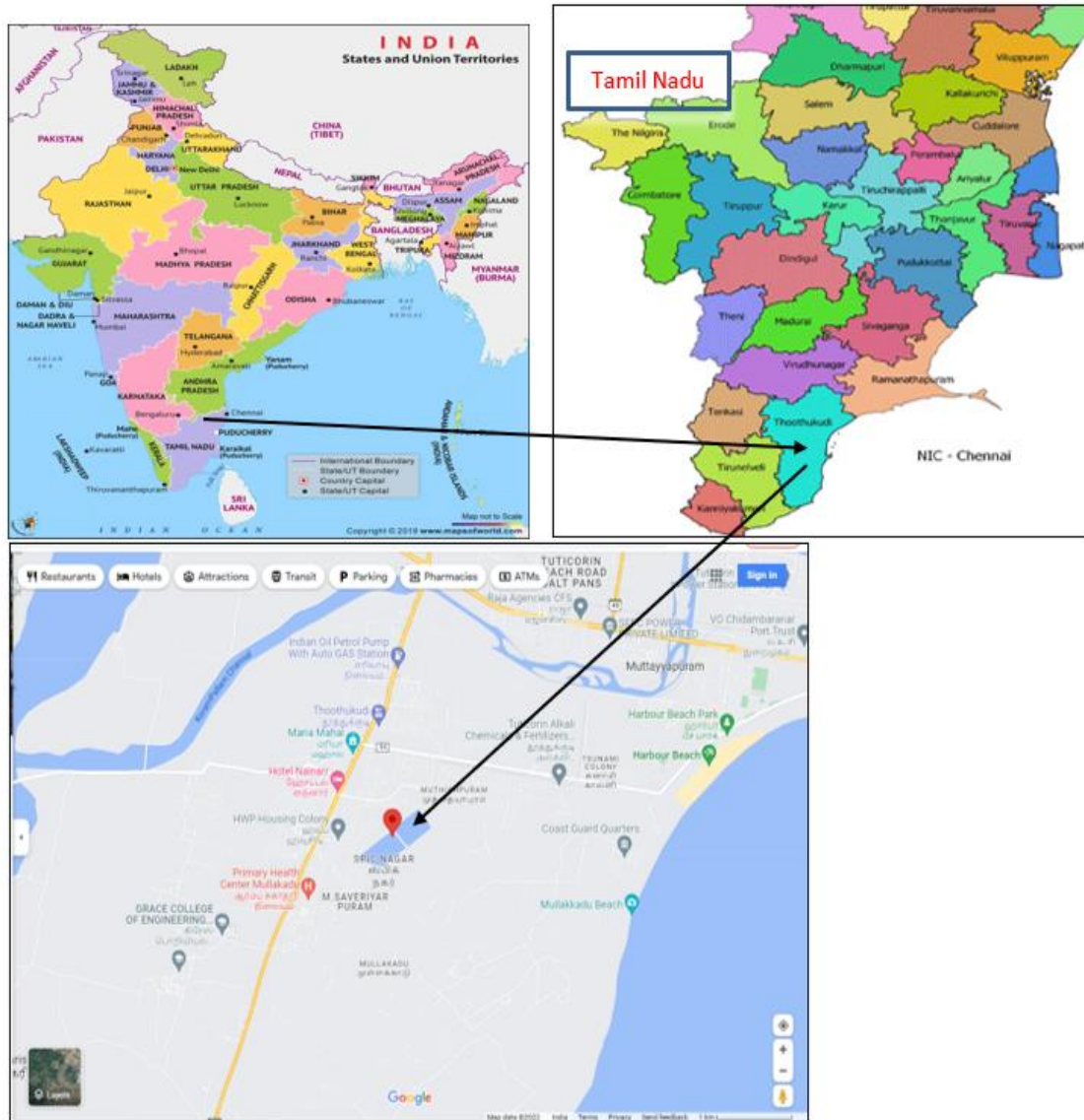


Figure 1: Location of the Project

1.13 Conditions Prior to Project Initiation

There was no project at the site of the project participant prior to the implementation of this project activity. The project participant was not involved in generation of solar PV based power and supplying it to the grid under the pre-project scenario, therefore, in the absence of the proposed project activity, the equivalent amount of electricity would have been generated from the connected / new power plants in the UNIFIED INDIAN GRID, while proposed project activity is a green field project. Therein, the main emission source in the pre-project scenario is the power plants connected to the UNIFIED INDIAN GRID and the primary GHG involved is CO₂

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Indian relevant laws and regulations. This project is voluntary initiatives by the Project Proponent.

- Electricity Act, 2003 and its subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003.
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020.
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021.
- Batteries (management and handling) rules 2019.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not participated in any other GHG Programs.

1.15.2 Projects Rejected by Other GHG Programs

The Project is not participated in any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits.

1.16.2 Other Forms of Environmental Credit

The Project has no intent to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable, as the project is not a supply chain project.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project achieves significant GHG emission reductions by harnessing the power from the Floating solar project. The project activity includes 22 MWac floating solar PV plant on the surface of two water reservoirs located near Muthiahpuram village, Thoothukudi district in the Indian state of Tamil Nadu. Thus, the project also provides long-term job opportunities to local people and contributes to the economic prosperity. Which directly contributes to the achieving following SDG's goals.

1.17.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
Sequential row number	SDG Target number	Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator	Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)	Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.	Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Towards addressing Quantity of net electricity supplied to the grid by project activity in year will replace equivalent amount of electricity feed to the grid by fossil fuel-based power plant.	The project activity includes the installation of 22 MW of renewable energy capacity that will deliver zero-emission electricity annually.	The project will Contribute renewable energy share in total grid energy consumption.
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Toward addressing the temporary and primary employment opportunity during the construction and operation phase.	Project has contributed directly to achieve the SDG target, because the project activity activity has created jobs in the renewable energy sector.	Which will also upgrade employee's knowledge and capacity over and above the commonly used technology in the energy sector of India.

3)	9.4	9.4.1 CO ₂ emission per unit of value added	GHG emission avoidance due to feeding of clean electricity generated from the project activity to the grid.	The project activity to result in avoiding annual CO ₂ equivalent emission avoidance of 36,873 tCO ₂ e /year	The project will Contribute in the lower down the emission
4)	13.2	13.2.2 Total greenhouse gas emissions per year	Implemented activities to result in avoidance of greenhouse Gas emission by refraining generation of electricity in fossil fuel -based power plant connected to the grid.	The project activity through implementation of 22 MW of floating Solar power generation unit has result in reduction of 36,873 tCO ₂ e /year	The project activity will result in the reduction of the total 258,117 tCO ₂ e throughout the crediting period.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

Further Information

There are no information or incidents that will have bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project 's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

There was no harm identified from the project and hence no mitigations measures are applicable.

2.2 Local Stakeholder Consultation

Local stakeholders were invited by sending invitation letter to the President of SoosaiNagar. The consultation was carried out on 11/03/2022 within plant premises. Total 22 nos. of Stakeholders consulted in the meeting.

The local stakeholder consultation process was carried out through a physical meeting at the project site. As a part of the consultation process, the project representatives explained all stakeholders about the type of development through the project and the technology used for power generation towards enhancing stakeholders' awareness about the project. During the meeting the project representative talked about the employment opportunities and other benefits likely to be rendered through this project and its contribution towards fighting against climate change.

The project owner also shared a feedback form during individual interviews during focused group discussions where the stakeholder can put down his/her complaint and the same if found genuine were to be addressed immediately. Also, regular stakeholder engagement is one the key focus at the site.

The project stakeholders were totally in support for setting up of these kinds of projects in the region.

The list of the Attendees is attached in the appendix 1.

2.3 Environmental Impact

Project activity is not having any negative environment Impact.

2.4 Public Comments

Not Applicable

2.5 AFOLU-Specific Safeguards

Not Applicable

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Methodology Title and reference: ACM0002 Grid-connected electricity generation from renewable sources --- Version 21.0⁶

The applied methodology draws upon the following tools:

Tool 01 - Tool for the demonstration and assessment of additionality (Version 07.0.0, EB 70, Annex8)⁷

Tool 07 - Tool to calculate the emission factor for an electricity system (Version 07.0, EB 100, Annex 4)⁸

Tool 24 – Common practice (Version 03.1, EB 84, Annex 7)⁹

Tool 27- Investment analysis (Version 12.0, EB 112 annex 2)¹⁰

⁶ <https://cdm.unfccc.int/methodologies/DB/HF3LP6O41YY0JIP1DK6ZRJO9RSCX3S>

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

⁸ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

3.2 Applicability of Methodology

The project activity is a green field grid connected captive solar power project with cumulative capacity of 22 MWac. Since the project is a large-scale project activity therefore Large-scale Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources applies to project activities that include retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant or construction and operation of a Greenfield power plant is applied. The applicability of the methodology is established through assessment of the following conditions.

Para No.	Applicability Conditions	Justification of eligibility
3	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity involves construction and operation of grid-connected captive Floating solar power (renewable energy) project at a site where no renewable energy power plant was operated prior to the implementation of the project activity and therefore a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (3.a). Hence the project activity meets the applicability condition of the methodology.
4	The methodology is applicable under the following conditions: (a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The project activity involves construction and operation of greenfield grid-connected captive Floating solar power project and hence complies to the applicability condition (4.a) of the methodology. Since the project activity does not include capacity additions, retrofits, rehabilitations, or replacements of existing plant/unit the applicability condition 4.b is not applicable/relevant for the project activity.

Para No.	Applicability Conditions	Justification of eligibility
5	In case of hydro power plants, one of the following conditions shall apply:¹¹ a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: (a) Lower than or equal to 15 MW; and (b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	The project activity involves construction and operation of greenfield grid-connected captive floating solar power project using solar energy for generation of electricity hence the applicability condition “5” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.
6	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction	The project activity involves construction and operation of greenfield grid-connected captive floating solar power project using solar energy for generation of electricity, hence the applicability condition “6” is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects.

¹¹ Project participants wishing to undertake a hydroelectric project activity that results in a new reservoir or an increase in the volume of an existing reservoir, where reservoirs have no significant vegetative biomass in the catchments area, may request a revision to the approved consolidated methodology.

Para No.	Applicability Conditions	Justification of eligibility
	of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
7	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity involves construction and operation of captive greenfield grid-connected floating solar power project using solar energy for generation of electricity hence the applicability condition 7" is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or biomass fired power plants/units.
8	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity involves construction and operation of captive greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition 8 is not relevant as the same pertains to retrofits, rehabilitations, replacements, or capacity additions.
9	In addition, the applicability conditions included in the tools referred to below apply	The applicability of the tools is outlined below

Applicability Conditions	Justification of eligibility
Tool 07: Tool to calculate the emission factor for an electricity system (Version 07.0)	
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This condition of applicable, OM, BM and CM are estimated using this tool (under section B.6.1) for calculating of the baseline emission.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter	Since the project activity is grid connected, the condition is applicable and emission

Applicability Conditions	Justification of eligibility
case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II.a and option II.b. If option II.a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	factor has been calculated accordingly.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a Non-Annex I country. Therefore, this condition is not applicable to the project activity.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project activity is a green field solar power plant and hence the condition of biofuel emission factor is not applicable.

Applicability Conditions	Justification of eligibility
Tool 01: Tool for the demonstration and assessment of additionality (version 07)	
Para 9 - The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project participant is not proposing new methodologies.
Para 10 - Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory	Project activity is applying additionality tool.

Applicability Conditions	Justification of eligibility
Tool 24: Common practice (Version 03.1)	

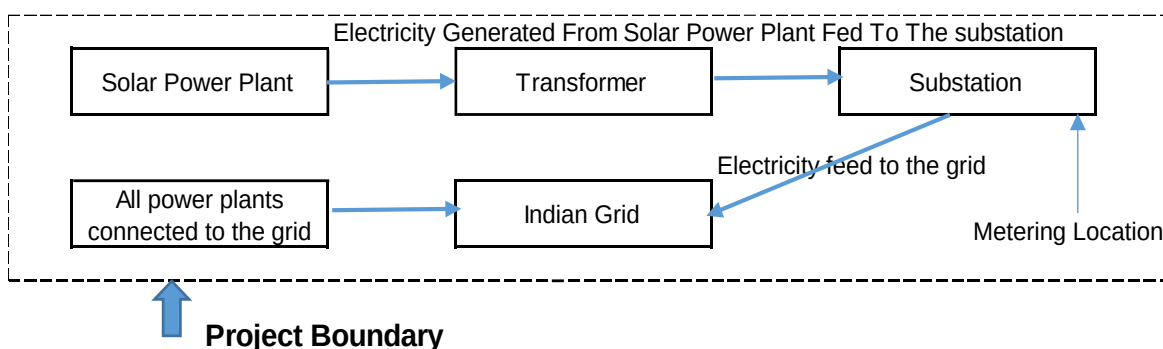
Applicability Conditions	Justification of eligibility
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality	The proposed project activity is applying the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied approved baseline and monitoring methodology does not define approaches for the conduction of the common practice test.

Applicability Conditions	Justification of eligibility
Tool 27: Investment analysis (Version 12.0)	
Para 2 - This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. – Proposed project activity is “Tool for the demonstration and assessment of additionality”	Proposed project activity applies Tool for the demonstration and assessment of additionality.
Para 3 - In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. – Methodology refers “General	The applied approved baseline and monitoring methodology contains requirements for the investment analysis that are not different from those

Applicability Conditions	Justification of eligibility
guidelines for SSC CDM methodologies, information on additionality (attachment A to Appendix B)	described in this methodological tool.

3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards the national grid of India.



The project boundary includes CO_{2e} as greenhouse gas (GHG) that would have emanated from the operation of grid connected fossil fuel-based power plants in the absence of the project activity.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main emission source
	CH ₄	No	Minor emission source
	N ₂ O	No	Not Applicable
Project	CO ₂	No	As a zero-emission grid connected Solar power project no emissions will result from the project activity in operation phase.
	CH ₄	No	As zero-emission grid connected

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Solar power project no emissions will result from the project activity.

3.4 Baseline Scenario

As per the approved consolidated Methodology ACM0002 (Version 21.0), baseline scenario would be defined as: “If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system (Version 07.0)”.

3.5 Additionality

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

- A Legal Requirement Test; and
- An Additionality Test either based on a Positive List test or a projects-specific additionality test.

The project is not mandated/enforced by law and is entirely a voluntary activity. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	This project applies an approved large-scale methodology ACM0002 “Consolidated baseline methodology for grid connected electricity generation from renewable sources”, Version 21.0. Selected methodology has been applied together with the “tool to calculate the emission factor for an electricity system, version 7.0” and “tool for assessment and demonstration of additionality, version 7.0.0”. These are the latest version of the methodology and related additionality & emission factor calculation tool.
Describe how the proposed project meets the criteria for deemed additionality.	Project revenue is not financially attractive as discussed in investment analysis section below (benchmark and sensitivity analysis). Continuation of the current situation supply of equal amount of electricity by the newly built grid connected power plants. Continuation of the current situation is not considered as a realistic alternative due to increasing electricity demand therefore

	new power plants should be constructed which includes mainly thermal power plants. Implementation of the project is additional to the baseline scenario which is an alternative 2 above and therefore reduces the emissions. The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate” for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations The Project activity conforms to all the applicable laws and regulations in India: 1. Power generation using renewable energy is not a legal requirement or a mandatory option¹². 2. There are state and sectoral policies, framed primarily to encourage renewable power projects. 3. These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments. 4. The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation. 5. There is no legal requirement on the choice of a particular technology for power generation The both alternatives are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity. In accordance with common practice analysis there is no plants similar to the proposed project and built without carbon revenue, the proposed type of project should not be considered as a common practice. Hence, project is additional in this aspect.
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Additionality Assessment for large scale project activity Instances

The table below is only applicable if the proposed project activity is a type of project activity which is deemed automatically additional, as defined by the applied approved methodology or standardized baseline.

¹²Electricity Act. 2003 and its subsequent amendments

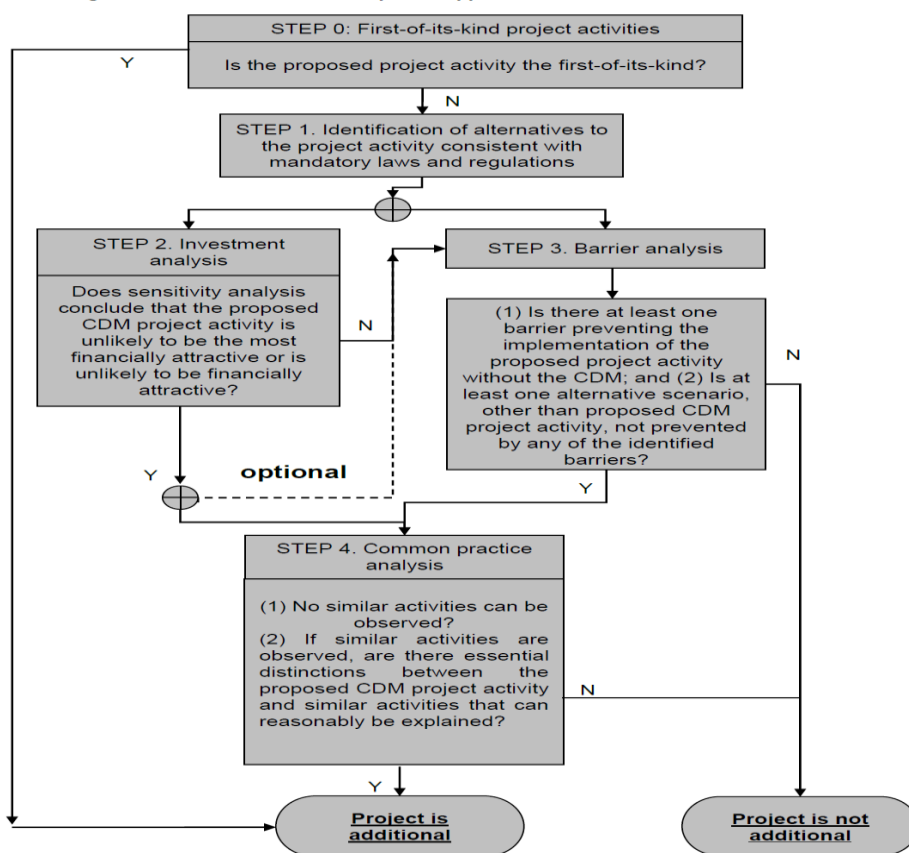
Specify the methodology or activity requirement or product requirement that establish deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	NA
Describe how the proposed project activity meets the criteria for automatic additionality in the relevant methodology or standardized baselines.	NA

The present project generates power using solar energy which is a renewable, zero emission source of energy. However, the project activity does not comply to the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies” and hence additionality of the project activity is demonstrated through a project specific additionality test.

Baseline considerations for the project are based on approved consolidated baseline methodology. ACM0002 (Version 21.0). The methodology requires the project investor to determine the additionality based on “Methodological Tool- Tool for the demonstration and assessment of additionality”, Version 7.0.0. The tool provides a step-wise approach to demonstrate and assess the additionality of a project (figure below). These steps are:

- (a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;
- (b) Step 1 Identification of alternatives to the project activity;
- (c) Step 2 Investment analysis;
- (d) Step 3 Barriers analysis; and
- (e) Step 4 Common practice analysis

Figure 1 Flowchart of the step-wise approach



The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The project activity is a large-scale solar power project undertaken in Indian state of Tamil Nadu. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.

The purpose of the project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and feed the electricity generated to the grid. Hence, the following alternatives are considered:

Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.

The option of proposed project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Fertilizer plant through India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of Plant and the country.

Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system Version 7.0"

Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.

Installation of a large-scale solar power project including the one proposed as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Tamil Nadu). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi supported by Central Pollution Control Board (CPCB). In accordance to

Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-july-2017 Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under “white category” for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate”. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations.

- Electricity Act, 2003 and it's subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003.
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021.
- E-waste (Management and Handling) Rules, 2020.
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021.
- Batteries (management and handling) rules 2019

Outcome of Sub-step 1b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment analysis

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in line with the Tools for sections as per “Investment Analysis” (Ver 12.0)

Sub-step 2a: Determine appropriate analysis method

There are three options for the determination of analysis method which are:

- I. Simple Cost Analysis
- II. Investment Comparison Analysis and
- III. Benchmark Analysis

The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the project owner was carried out before the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

According to the “Tool for demonstration and assessment of Additionality”, Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 12¹³.0 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = 10.55%

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 10.55% (as per Appendix of CDM TOOL 27: Investment Analysis)

Inflation Rate forecast for by Reserve Bank of India (RBI) (i.e. Central Bank of India) for India.

Benchmark estimation:

Project	Date of Investment Decision	Commissioning date
Floating Solar project of Greenam Energy	23/01/2018	12/01/2022

Inflation Rate (at the time of investment decision making)

Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products for 10 years)	Document Date	Sources Link
4.0%	06/04/2017	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458

Benchmark for Floating Solar project of Greenam Energy

Description	Value	Sources Link
Default Value for India as per UNFCCC guidelines	10.55%	EB 112, Annex 2, TOOL27 "Investment Analysis", Version 11.0 [web link: https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf]
Inflation forecast (WPI Mean of annual forecasts of Non-food Manufactured Products)	4.0%	(Table. A.9 – Annual average percentage change) https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=17458
Benchmark	14.97%	Calculated

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III): The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity. substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

The input assumption referred below:

Technical Parameters			
Particulars	Value	Unit	Source
Capacity of the plant	22	MW	DPR
Plant Load factor	21.85%	%	DPR
Grid Availability	100	%	As a conservative approach
Auxiliary Consumption	0.4%	%	DPR
Transmission Loss & Charges	3.91%	%	as per TNERC Order TP No:2 of 2017 dt 11.08.2017
Annual Module degradation	0.43%	%	DPR
Working days in a year	365	Days	DPR
Working hours in a day	24	Hrs	DPR
Technical Lifetime of the project	25	Years	DPR

Financial Parameters			
Particulars	Value	Unit	
Annual O & M cost	7.4	Million INR/annum	DPR
Escalation in O&M	5.00%	%	DPR
Insurance	0.09%	%	DPR
Admin Expenses	2.86	Million INR/annum	DPR
Escalation in Admin Expenses from 2nd year onwards	5.00%	%	DPR

Inverter warranty	0.78	Million	DPR
Project cost	1274	Million INR	DPR
Equity @ 30% of total cost	382.2	Million INR	Management decision
Residual/scrap/salvage Value	10%	%	
Tariff rate	4.12	INR/ kWh	Reverse bid order
Depreciable amount	85%	%	Calculated Value
Depreciation (Book depreciation) on civil works	10%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/
Depreciation (Book depreciation) on plant & machineries	8%	%	https://taxadda.com/depreciation-rates-as-per-companies-act-2013/
IT depreciation on building & civil works	7.69%	%	IT depreciation rate ¹⁴
IT depreciation on Plant & Machineries	7.69%	%	IT depreciation rate ¹⁵
Corporate Tax Rate	25.17%	%	https://www.pwc.com/mu/en/services/india-desk/corporate-tax.html
Interest rate on loan	10.5%	%	DPR
Commercial operation date	12/01/2022		Commissioning certificate

Outcome of Step 2:

Calculation of equity IRR is submitted in the form of excel sheet.

Equity IRR	Benchmark (Equity IRR)
7.50%	14.97%

¹⁴ <https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69>

¹⁵ <https://search.incometaxindia.gov.in/pages/results.aspx?k=7%2E69>

This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

Sub-step 2d: Sensitivity Analysis

Addressing Guidance under Sensitivity Analysis Version-11.0, following factors has been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The rationale of sensitivity is, "The ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis."

Sensitivity Analysis

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the project cost, O&M cost, PLF and Tariff Rate Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

Probability to breach the benchmark:
Sensitivity Parameter 1 : PLF or generation
PLF considered in financials for is as per Third Party report in line with "Guidelines for the reporting and validation of Plant load factors" stated in EB 48 Annex11 option 3(b). Hence, variation in PLF of more than 10% is unlikely to happen as the PLF has been reported as per the Third Party Report based on long term data. IRR is not crossing the benchmark if PLF increases by 10% in any circumstances.
Sensitivity Parameter 2: O&M
The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely.
Sensitivity Parameter 3 : Project Cost
Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range
Sensitivity Parameter 4 : Tariff Rate

The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached.

Result of Sensitivity Analysis Table:

Parameters	-10%	IRR (Base Case)	10%
PLF or Generation	4.08%	7.50%	10.84%
O&M cost	7.74%	7.50%	7.26%
Project Cost	10.82%	7.50%	4.77%
Tariff	4.08%	7.50%	10.84%

Step 3: Barrier Analysis

As per Tool for demonstration and assessment of additionality” (Version 07.0.0), Step 2 or Step 3 or both can be used to demonstrate additionality of the project activity. In this case, Step 3 is not being used for the purpose.

Step 4: Common Practice Analysis

As per para 57 of Tool for demonstration and assessment of additionality” (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 22 MW Floating PV Solar Power Project at Tamil Nadu by Greenam Energy pvt Ltd

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity	Unit
+50%	33	MW
Capacity of the proposed project activity	22	MW
-50%	11	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;

- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity range for the chosen projects.
- The projects started commercial operation before the PSF is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- As the project is located in Tamil Nadu state of India, therefore, projects in the geographical area Tamil Nadu have been chosen for analysis. The project activity involves generation of electricity from solar energy. The project activity is located in the states of Tamil Nadu in India and the policy applicable for the solar projects is regulated by respective state policy. The policies/tariff for each state is regulated by State Electricity Regulatory Commissions of respective states and they differ for respective states. The project is Floating PV solar and a 30% equity-based project and generated power will be used for captive purpose through grid and paying wheeling charges. Hence, all state electricity regulation for such projects only applicable for this project. So, solar projects, which are exporting electricity to grid are followed different state electricity regulatory policy.
- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.
- The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 11 MW to 33 MW.
- Purchase Order date of the project activity is 28/08/2018, therefore, projects, which have started commercial operation before 28/08/2018, have been considered for analysis.

Numbers of similar projects identified, which fulfil above-mentioned conditioned are N solar = 0

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, Nall.

Project activities, which have got registered or are under validation with CDM/VCS/GS/GCC have been excluded in this step. The list of the power plants identified is provided to the Verifier. After excluding the registered and under validation projects the total number of projects.

$N_{all} = 0$

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

As per the tool on Common Practice, the project activities have been separated from the different technologies on the basis two criteria:

Different technologies - are technologies that deliver the same output and differ by at least one of the following (as appropriate in the context of the measure applied in the proposed project activity and applicable geographical area):

(a) Energy source/fuel (example: energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas);

(b) Feed stock (example: production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of alternative fuels or less carbon-intensive fuels);

(c) Size of installation (power capacity)/energy savings:

(i) Micro (as defined in paragraph 24 of decision 2/CMP.5 and paragraph 39 of decision 3/CMP.6); (ii) Small (as defined in paragraph 28 of decision 1/CMP.2);

(iii) Large.

(d) Investment climate on the date of the investment decision, inter alia:

(i) Access to technology;

(ii) Subsidies or other financial flows;

(iii) Promotional policies;

(iv) Legal regulations.

(e) Other features, inter alia:

(i) Nature of the investment (example: unit cost of capacity or output is considered different if the costs differ by at least 20 %).

The Project is 30% equity-based and Floating PV solar Captive project and deployed for export of power after paying wheeling charges. Hence, the investment climate of the project is different from any other project of different tariff or revenue structure. So solar power project that are not covered under reverse bid project are considered as different from the project activity.

Hence, projects where either of the conditions is satisfied those projects are counted for calculating

N_{diff} projects.

$N_{diff} = 0$

Step (5): calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology

used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor for the project activity is calculated as follows:

In terms of the formula: $F = 1 - N_{diff} / N_{all}$, the result is mathematically meaningless.

$N_{all} - N_{diff} = 0$

Thus, the project does not fulfil the requirement that F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3, and the project is not a common practice

3.6 Methodology Deviations

This section is not applicable, as there is no deviation of methodology applied by the Project.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Not Applicable

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The baseline emissions are the product of generate power from the solar power plant baseline $EG_{PJ,y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor. As per Methodology ACM0002 Ver-21.0 the baseline emission equations are used.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad \text{Equation (1)}$$

Where:

BE_y	=	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

For Green field Project

$$EG_{PJ,y} = EG_{facility,y} \quad \text{Equation (2)}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Annual Baseline emission for first year of crediting period

Capacity (MW)	PLF (%)	Auxiliary Consumption	Transmission loss	Net Generation	Baseline emissions factor	Baseline emissions	Emission reductions
MW	%	%	%	(MWh/year)	(tCO ₂ /MWh)	(tCO ₂ /year)	(tCO ₂ /year)
22	21.85	0.4	3.91	40,294	0.9310	37,514	37,514

5.2 Project Emissions

As per para 31 of the applied methodology ACM0002 (version 21.0), no project emissions are considered in this project activity.

Hence, project emissions PE_y = 0 tCO₂e

5.3 Leakage

As per methodology ACM0002 v21, Leakage emission as per methodology is “The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected”.

5.4 Estimated Net GHG Emission Reductions and Removals

The annual emission reductions ER_y for the project activity are calculated as the baseline emissions minus the project emissions. Hence the Project emission and Leakage emission is 0. Therefore, ER_y = BE_y

Year	Estimated baseline emissions or	Estimated project emissions or	Estimated leakage	Estimated net GHG emission reductions or
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	removals (tCO ₂ e)	removals (tCO ₂ e)	emissions (tCO ₂ e)	removals (tCO ₂ e)
Year 1	37,352	0	0	37,352
Year 2	37,192	0	0	37,192
Year 3	37,032	0	0	37,032
Year 4	36,872	0	0	36,872
Year 5	36,714	0	0	36,714
Year 6	36,556	0	0	36,556
Year 7	36,399	0	0	36,399
Total	258,117	0	0	258,117

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid, OM, y}
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor in the year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9518
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation

Comments	This parameter is fixed ex-ante for the entire crediting period.
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Data / Parameter	$EF_{grid, BM, y}$
Data unit	tCO ₂ /MWh
Description	Build Margin CO ₂ emission factor in the year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.869
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid, CM, y}$
Data unit	tCO ₂ /MWh
Description	Combine Margin CO ₂ emission factor in the year y
Source of data	CO ₂ Emission Database, Version 18.0, December 2022 published by Central Electricity Authority (CEA), Government of India.
Value applied:	0.9310
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable
Purpose of Data	For baseline emissions calculation

Comments

This parameter is fixed ex-ante for the entire crediting period.

6.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ, y}																																		
Data unit	MWh/year																																		
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																																		
Source of data	Monthly Joint Meter reading																																		
Description of measurement methods and procedures applied	Not Applicable																																		
Frequency of monitoring/recording	Continuous measurement																																		
Value applied:	27,724 MWh (Average annual net generation for 7 years)																																		
Monitoring equipment	<table border="1"> <tr> <td colspan="2">Main meter</td></tr> <tr> <td>Type of meter(s)</td><td>E3M024, 3Ph, 4wire</td></tr> <tr> <td>Location of meter(s)</td><td>At plant switch yard.</td></tr> <tr> <td>Accuracy of meter(s)</td><td>0.2S</td></tr> <tr> <td>Serial number of meter(s)</td><td>TNW02796 (Meters are subjected to replaced or changed as per requirement).</td></tr> <tr> <td>Calibration frequency</td><td>Once in five years</td></tr> <tr> <td>Date of Calibration/ validity</td><td>26/10/2021</td></tr> <tr> <td>Reference No. of Calibration Certificates</td><td>GEPC/2021-22/310/CC/2774</td></tr> <tr> <td>Calibration Status</td><td>Ok</td></tr> <tr> <td colspan="2">Check meter</td></tr> <tr> <td>Type of meter(s)</td><td>E3M024, 3Ph, 4wire</td></tr> <tr> <td>Location of meter(s)</td><td>At plant switch yard.</td></tr> <tr> <td>Accuracy of meter(s)</td><td>0.2S</td></tr> <tr> <td>Serial number of meter(s)</td><td>TNW02797 (Meters are subjected to replaced or changed as per requirement).</td></tr> <tr> <td>Calibration frequency</td><td>Once in five years</td></tr> <tr> <td>Date of Calibration/ validity</td><td>26/10/2021</td></tr> <tr> <td>Reference No. of Calibration Certificates</td><td>GEPC/2021-22/310/CC/2775</td></tr> </table>	Main meter		Type of meter(s)	E3M024, 3Ph, 4wire	Location of meter(s)	At plant switch yard.	Accuracy of meter(s)	0.2S	Serial number of meter(s)	TNW02796 (Meters are subjected to replaced or changed as per requirement).	Calibration frequency	Once in five years	Date of Calibration/ validity	26/10/2021	Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2774	Calibration Status	Ok	Check meter		Type of meter(s)	E3M024, 3Ph, 4wire	Location of meter(s)	At plant switch yard.	Accuracy of meter(s)	0.2S	Serial number of meter(s)	TNW02797 (Meters are subjected to replaced or changed as per requirement).	Calibration frequency	Once in five years	Date of Calibration/ validity	26/10/2021	Reference No. of Calibration Certificates	GEPC/2021-22/310/CC/2775
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	Calibration Status	Ok
QA/QC procedures applied	The meters are approved, tested & sealed by the State Utility and are in the custody of State Utility. The metering arrangement, accuracy class of meters, calibration frequency is under control of state utility and Project owner do not have any control on it. The calibration of all the meters is planned to be is carried out in-line with the National standard which recommends at least once in 5-year calibration. Faulty meters will be duly replaced. The meters are of accuracy class 0.2s - Tariff metering point is consisting of Main & Check meters for both Import & Export. - Meter will reset automatically at 00:00 hrs on 1st of every month and it is AMR (Automatic Meter Reading) reading for both Import & Export. So apportioning is not required. - All the reading values are stored in the location of "Billing". - The stored readings are transferred to Main server from Tariff meter thro' Modem. - The stored readings will be taken by TNEB-Tangedco officials manually on 1st of every month from the meters storage location (Billing). - From the Server the AMR readings & Generation statement will be generated and will be available in the OA portal of the consumer in Tangedco website and it can be downloaded In absence or delay in the meter calibration appropriate Guidelines will be applied appropriately to confirm the conservativeness of metering.	
Purpose of data	To calculate baseline emission	
Calculation method	The Net electricity exported/supplied to the grid by the project activity is estimated as a difference of electricity exported to the grid and electricity imported from the grid. The net electricity supplied/exported to the grid can be obtained from the record of Monthly Joint Meter Reading by the state utility	
Comments	Data will be archived in paper & electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.	

6.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for CDM project activities and is proposed for grid-connected solar power project being implemented. The monitoring plan, which will be implemented by the project owner describes about the monitoring organisation, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

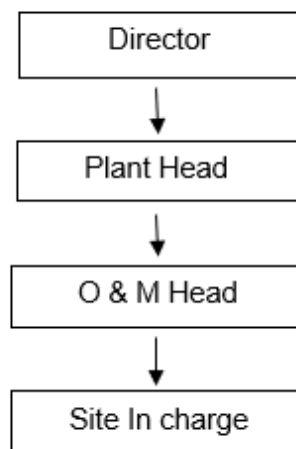
The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project participant. PO proposed the following structure for data monitoring, collection, data archiving and calibration of equipment's for this project activity. The team comprises of the following members:

Organizational Structure for Monitoring

Responsibilities of Head (Director): Tracking and reviewing the overall functioning and maintenance of the project activity from Plant Head (Operations). Plant Head (Operations) will be reporting Director.

Responsibilities of Plant Head : Overall functioning of the project activity and Coordinating with the O & M Manger and Team for the proper functioning of Project activity. He will be reporting to Director.

Responsibilities of O & M Head: O & M Manager and team is responsible for Operations and Maintenance related issues, they are also responsible for day-to-day data collection and monitoring, ensures completeness and reliability of data (calibration of equipment).



The Site In-charge will be responsible for carrying out internal auditing and QA/QC. All the values from generation record will be checked with JMR and invoices for consistency. In case there are any non-conformances identified. The Site In-charge will investigate the error and revise the record to correct it. In any case where values have slightest of variation in different records the most conservative value will be taken in the project monitoring report.

Data Measurement

The export and import energy will be measured continuously using Main and Check meters located at the substations. Readings of meters shall be taken on monthly basis by authorized officer of utility in the presence of the representative of Project Owner (PO). Based on the Meter Reading Statement, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity.

Data Collection and Achieving

Readings from meters will be collected in the presence of the plant in-charge. Export and Import data would be recorded and stored in logs as well as in electronic form. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of ACCs for the project activity whichever occurs later.

Emergency Preparedness

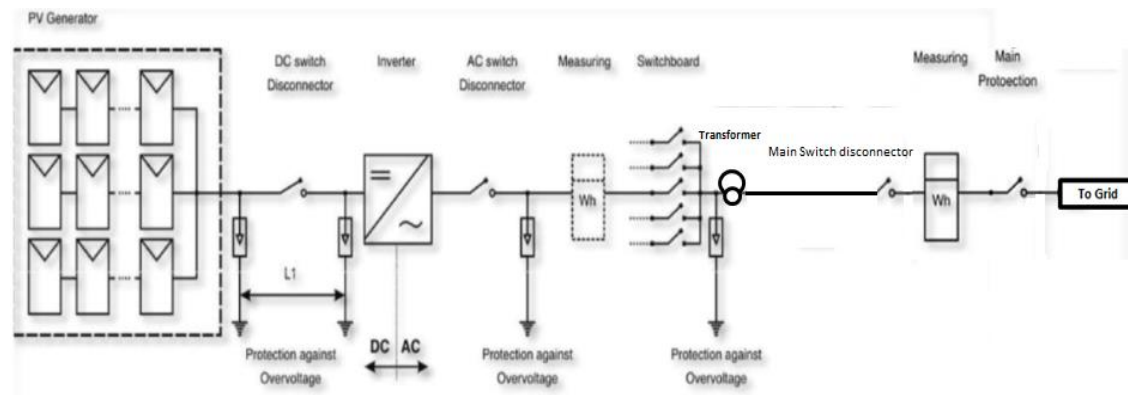
The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the event that the main meter, which is used to record the net electricity exported by the project, is found to be faulty it will be repaired or replaced by authorized officer of SEB and the data from the check meter will be used in its place. In the unlikely event that the check meter fails it will also be repaired or replaced.

Personnel Training

In order to ensure proper functioning of the project activity and proper monitoring of emission reductions, the staff will be trained. The plant staff will be trained in equipment operation, data recording, reports writing, operation & maintenance, and emergency procedures in compliance with the monitoring plan.

Metering Arrangement

Line diagram with metering arrangement for the solar project activity is shown below:



QA/QC Procedures

The energy meters at the feeders are maintained and owned by state electricity board. Neither the project owner nor the site personnel have any control over it. The records will be crosschecked

with the records of sold electricity. The meters are calibrated by state electricity board out in-line with the Nation standard which recommends at least once in 5-year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Value applied:	27,724
Comments	-

7.2 Baseline Emissions

The baseline emissions is the product of net electricity generation supplied to the grid expressed in MWh by the project multiplied by an emission factor.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad \text{Equation (3)}$$

Where:

- BE_y = Baseline emissions in year y (t CO₂/yr)
- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
- $EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO₂/MWh)

$$\begin{aligned}
 BE_y &= 27,724 \times 0.9310 \\
 &= 25,811
 \end{aligned}$$

7.3 Project Emissions

The project is a solar farm, belongs to renewable energy activity, hence $PE_y = 0 \text{ tCO}_2\text{e/MWh}$.

7.4 Leakage

No leakage is considered for this project.

7.5 Net GHG Emission Reductions and Removals

$$ER_y = BE_y - PE_y$$

Where:

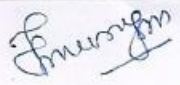
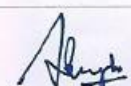
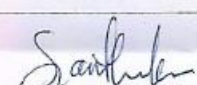
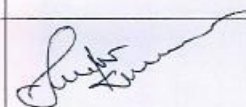
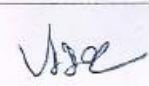
ER_y = Emission reductions in year y (tCO_2)

BE_y = Baseline Emission in year y (tCO_2)

PE_y = Project Emission in year y (tCO_2)

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
Year 1	37,352	0	0	37,352
Year 2	37,192	0	0	37,192
Year 3	37,032	0	0	37,032
Year 4	36,872	0	0	36,872
Year 5	36,714	0	0	36,714
Year 6	36,556	0	0	36,556
Year 7	36,399	0	0	36,399
Total	258,117	0	0	258,117

APPENDIX 1: LIST OF ATTENDEES

ATTENDANCE SHEET			
STAKEHOLDERS CONSULTATION MEETING			
GCC Project		: Floating Solar project of Greenam Energy	
Organizer		: Greenam	
Project Location		: Tuticorin	
VENUE: Greenam office, SPIC Nagar		DATE: 11/03/2022	
Sl. No.	Name	Designation	Signature
1.	Arumugam.M	Sreesainagar	
2.	Rajesh.P	"	
3.	Santhana kumar	"	
4.	Siva kumar	"	
5.	Vijayan	"	
6.	Jayavel	"	
7.	Venkatesan	"	

ATTENDANCE SHEET

STAKEHOLDERS CONSULTATION MEETING

GCC Project : Floating Solar project of Greenam Energy

Organizer : Greenam

Project Location : Tuticorin

VENUE: Greenam office,
SPIC Nagar

DATE: 11/03/2022

8.	Ganesan	Sasarnagar	[Signature]
9.	Rajesh	"	[Signature]
10.	Isravel	"	[Signature]
11.	Gomathinayagam	Thangamalai	[Signature]
12.	Suresh Kumar	"	[Signature]
13.	Thiyagarajan	"	[Signature]
14.	Melvin	"	[Signature]
15.	Palavesam	"	[Signature]

ATTENDANCE SHEET

STAKEHOLDERS CONSULTATION MEETING

GCC Project : Floating Solar project of Greenam Energy

Organizer : Greenam

Project Location : Tuticorin

VENUE: Greenam office,
SPIC Nagar

DATE: 11/03/2022

16.	Umashankar	Sosainagar	Umashankar D.
17.	Dinesh	"	Dinesh Kumar
18.	Vaanaamaalai	"	S. Vaanaamani
19.	Murugan	"	Murugan
20.	Rajagopalan	"	Rajagopalan
21.	Jhonson	"	Jhonson
22.	Perumal	"	Perumal